Counterexamples to a conjectured characterization of cubes by the lattice Borsuk number, formally verified in Lean 4

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Abstract

For a bounded set $S \subset \mathbb{Z}^d$, the lattice Borsuk number $\beta_{\mathbb{Z}}(S)$ is the least number of parts in a partition of S into parts of strictly smaller lattice diameter. Brose, De Loera, Lopez-Campos, and Torres proved $\beta_{\mathbb{Z}}(S) \le 2^d$ and conjectured (arXiv:2508.20009, Conjecture 3) that $\beta_{\mathbb{Z}}(S) = 2^d$ if and only if $\mathrm{conv}(S)$ is unimodularly equivalent to a cube $[0, m]^d$. We show that the conjecture is false as stated: the four-point set $S_A = \{(0, 0), (1, 0), (0, 1), (3, 5)\} \subset \mathbb{Z}^2$ has all six pairwise differences primitive, hence $\beta_{\mathbb{Z}}(S_A) = 4 = 2^2$, while $\mathrm{conv}(S_A)$ contains exactly 7 lattice points and so is unimodularly equivalent to no square: the count is a unimodular invariant and 7 is not a perfect square. This disproof is formally verified: a Lean 4 development of about 1100 lines over mathlib proves `betaZ SA = 4`, `latticeCount hullSA = 7`, and `conjecture3_false`, with the formal definitions documented against the verbatim text of the paper; the axiom audit reports only the three standard axioms. Two further counterexamples, verified in exact arithmetic but not formalized, close the natural repair: a four-point set $T = \mathrm{conv}(T) \cap \mathbb{Z}^2$ whose hull is a triangle of area 3/2, and an eight-point set $C = \mathrm{conv}(C) \cap \mathbb{Z}^3$ with $\beta_{\mathbb{Z}}(C) = 8 = 2^3$ whose hull has volume 5/2. The conjecture therefore fails even when restricted to full sets of lattice points of lattice polytopes, in dimensions 2 and 3.

1 Introduction

In a recent study of lattice diameters, Brose, De Loera, Lopez-Campos, and Torres [3] introduced a discrete analogue of Borsuk's classical partition problem [2, 4]. Throughout, $d \ge 1$ and all notions are those of [3]. For $x, y \in \mathbb{Z}^d$ write $[x, y] := \mathrm{conv}(\{x, y\})$ and

$$f(x, y) \;=\; \big|[x, y] \cap \mathbb{Z}^d\big| - 1,$$

the number of lattice points on the segment, minus one. For a bounded (equivalently, finite) set $S \subset \mathbb{Z}^d$ the *lattice diameter* is $\mathrm{diam}_{\mathbb{Z}}(S) = \max_{x,y \in S} f(x, y)$, and the *lattice Borsuk number* $\beta_{\mathbb{Z}}(S)$ is the minimal size of a partition of S in which every part has lattice diameter strictly smaller than $\mathrm{diam}_{\mathbb{Z}}(S)$ [3, Definition 1.3 and §4.2]. A vector $u \in \mathbb{Z}^d$ is *primitive* if the gcd of its coordinates is 1; two sets $P, Q \subseteq \mathbb{R}^d$ are *unimodularly equivalent* if $P = AQ + t$ for some $A \in \mathrm{GL}(d, \mathbb{Z})$ and $t \in \mathbb{Z}^d$ (the paper states this notion for lattice polytopes; every set to which we apply it below is one — see Section 4).

Theorem 4.8 of [3] solves the discrete Borsuk partition problem: $\beta_{\mathbb{Z}}(S) \le 2^d$ for every bounded $S \subset \mathbb{Z}^d$, with equality for $S = \{0, 1\}^d$; the proof bounds the maximum degree of the *lattice*

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Borsuk graph (vertex set S, edges the pairs realizing $\operatorname{diam}_{\mathbb{Z}}(S)$) by $2^d - 1$ via a collinearity lemma of Rabinowitz [7] and applies Brooks' theorem, using that $\beta_{\mathbb{Z}}(S)$ is the chromatic number of this graph. The paper closes with a conjectured characterization of the equality case, which we quote verbatim (Conjecture 3 of [3], §4.2):

> *"Let $S \subset \mathbb{Z}^d$ be a bounded set. Then $\beta_{\mathbb{Z}}(S) = 2^d$ if and only if* $\operatorname{conv}(S)$ *is unimodularly equivalent to a d-cube $[0, m]^d$ for any $m \in \mathbb{N}$."*

(The phrase "for any $m \in \mathbb{N}$" is read as "for some $m \in \mathbb{N}$"; our counterexamples refute the equivalence under either reading, and for $\mathbb{N}$ with or without 0 — see Section 4.) This note proves:

Theorem 1.1 (formally verified; see Theorem 2.3 and Section 4)**.** *Conjecture 3 of [3] is false as stated: its $d = 2$ instance fails. The set $S_A = \{(0, 0), (1, 0), (0, 1), (3, 5)\} \subset \mathbb{Z}^2$ satisfies $\beta_{\mathbb{Z}}(S_A) = 4 = 2^2$, yet* $\operatorname{conv}(S_A)$ *is unimodularly equivalent to no square* $[0, m]^2$*,* $m \in \mathbb{N}$.

Theorem 1.2 (exact arithmetic, not formalized; see Theorems 3.1 and 3.2)**.** *The conjecture fails even under the additional hypothesis* $S = \operatorname{conv}(S) \cap \mathbb{Z}^d$*, in dimensions* 2 *and* 3*: the sets*

$$T = \{(0, 1), (1, 0), (1, 1), (2, 2)\} \subset \mathbb{Z}^2 \quad \textit{and}$$
$$C = \{(0, 2, 2), (1, 0, 1), (1, 1, 2), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (3, 1, 3)\} \subset \mathbb{Z}^3$$

are the full lattice-point sets of their hulls, attain $\beta_{\mathbb{Z}} = 2^d$, and their hulls are unimodularly equivalent to no cube.

In all three witnesses it is the "only if" direction of the conjecture that fails; the upper bound $\beta_{\mathbb{Z}}(S) \leq 2^d$ of [3, Theorem 4.8] is of course untouched. The mechanism is the degenerate regime $\operatorname{diam}_{\mathbb{Z}}(S) = 1$: there every part of an admissible partition must be a singleton, so $\beta_{\mathbb{Z}}(S) = |S|$, and $\beta_{\mathbb{Z}}(S) = 2^d$ merely says that S consists of 2^d points with all pairwise differences primitive — a condition that places almost no constraint on the shape of $\operatorname{conv}(S)$. The sharpness example $\{0, 1\}^d$ of [3, Theorem 4.8] itself lives in this regime, and the lattice Borsuk graph of each of our witnesses is the complete graph K_{2^d}, exactly the situation the conjecture was designed to characterize; the failure is therefore not an artifact of a perverse reading. Theorem 1.2 matters because the authors remark [3, §4.2] that in dimension two the conjecture "is true and can be proven using a characterization of lattice polygons which have four diameter directions" of Bárány and Füredi [1, Theorem 2(iii)] — a statement about lattice polygons, i.e. about sets of the form $\operatorname{conv}(S) \cap \mathbb{Z}^2$. Witness T shows that no reading of "bounded set" as "full lattice-point set of a polygon" can save the $d = 2$ claim: any correct repair must also exclude the diameter-one regime.

What is and is not machine-checked deserves precision. Theorem 1.1 is proved in Lean 4 [6] over mathlib [5]: the statement of the conjecture is formalized in `Borsuk/Defs.lean` with every definition documented against the verbatim text of [3], and the refuting theorem `conjecture3_false` is checked by the Lean kernel, with an axiom audit reporting only `propext`, `Classical.choice`, and `Quot.sound` (Section 4). Theorem 1.2 is *not* formalized; it is verified by exact integer and rational arithmetic (no floating point) in a short self-contained script distributed with the artifact and re-run while preparing this note. A literature check (June 2026, most recently on 2026-06-12) found no prior or concurrent resolution of the conjecture; arXiv:2508.20009 remains at version 1.

Methods and AI-use disclosure. The counterexamples and the formal proof were produced by Demonstrandum, a verification-first multi-agent AI workflow built on Claude language models (Anthropic). Agents froze the conjecture statement from the arXiv v1 LaTeX source, constructed witness A and three independent $d = 3$ full-set witnesses (of which C is one) by direct search with exact verification, and wrote the Lean development in four work packages against frozen formal definitions. An agent from a different model family (Codex, GPT-5.5, OpenAI) independently re-verified all witness computations, contributed witness T, searched independently for prior resolutions, and adversarially reviewed the faithfulness of the formal definitions against the paper text. The Lean kernel — not any language model — is the guarantor of every formalized claim; faithfulness of the formal *definitions* to the paper's statement is the one link the kernel cannot check, and Section 4 documents how it was controlled. The text of this note was likewise drafted with the Claude-based pipeline; the author directed the work and takes responsibility for its content. Every number appearing in this note was recomputed during its preparation (June 2026): the Lean project was rebuilt and its axiom audit re-run, and the witness data were re-verified by the exact-arithmetic scripts `_recompute.py` and `_referee_recheck.py` included with the artifact (Section 5).

2 Witness A and the disproof

The two lemmas below are stated for general d; the Lean formalization instantiates them at $d = 2$ (Section 4), which suffices, since Conjecture 3 asserts all dimensions at once and witness C is handled separately. For $v \in \mathbb{Z}^d$ let $\gcd(v)$ denote the gcd of the coordinates of v, with $\gcd(0) = 0$.

Lemma 2.1 (segment count; `Borsuk.ncard_segLatticePts`). *For $x, y \in \mathbb{Z}^d$ with $x \neq y$, let $g = \gcd(y - x) \geq 1$ and $u = (y - x)/g$. Then*

$$[x, y] \cap \mathbb{Z}^d \; = \; \{\, x + ku : k = 0, 1, \ldots, g \,\},$$

so $|[x, y] \cap \mathbb{Z}^d| = g + 1$; for $x = y$ the segment is the single lattice point x and $\gcd(0) = 0$, so in all cases $f(x, y) = \gcd(y - x)$. In particular $f(x, y) = 1$ exactly when $y - x$ is primitive.

Proof. The vector u is primitive and $x + ku = (1 - \frac{k}{g})x + \frac{k}{g}\, y \in [x, y]$ for $0 \leq k \leq g$, giving $\supseteq$. Conversely, a point $p \in [x, y]$ has the form $p = x + t(y - x) = x + (tg)u$ with $t \in [0, 1]$. By Bézout there are integers $\alpha_1, \ldots, \alpha_d$ with $\sum_i \alpha_i u_i = 1$; applying them to the integer vector $p - x = (tg)u$ gives $tg = \sum_i \alpha_i (tg) u_i \in \mathbb{Z}$, and $0 \leq tg \leq g$, so $p = x + ku$ with $k = tg \in \{0, \ldots, g\}$. The map $k \mapsto x + ku$ is injective ($u \neq 0$), so the count is $g + 1$. □

Lemma 2.2 (diameter-one sets; `Borsuk.betaZ_eq_card_of_latticeDiam_eq_one`). *Let $S \subset \mathbb{Z}^d$ be finite with $|S| \geq 2$ and all pairwise differences of distinct points primitive. Then $\operatorname{diam}_{\mathbb{Z}}(S) = 1$ and $\beta_{\mathbb{Z}}(S) = |S|$.*

Proof. By Lemma 2.1, $f(x, y) = 1$ for all distinct $x, y \in S$, so $\operatorname{diam}_{\mathbb{Z}}(S) = 1$. In a partition admissible for $\beta_{\mathbb{Z}}$, every part has lattice diameter < 1, i.e. 0; but a part containing distinct points $x \neq y$ has diameter at least $f(x, y) \geq 1$, since distinct lattice points always have $\gcd(y - x) \geq 1$. Hence all parts are singletons and there are at least $|S|$ of them; the partition into singletons is admissible ($0 < 1$), so $\beta_{\mathbb{Z}}(S) = |S|$. □

Theorem 2.3 (formalized: `witnessA_kills_conjecture3`, `conjecture3_false`). *Let $S_A = \{(0, 0),\, (1, 0),\, (0, 1),\, (3, 5)\} \subset \mathbb{Z}^2$. Then:*

(i) $\beta_{\mathbb{Z}}(S_A) = 4 = 2^2$*;*

(ii) $\operatorname{conv}(S_A) \cap \mathbb{Z}^2$ *consists of exactly the* 7 *points* $(0,0)$*,* $(1,0)$*,* $(0,1)$*,* $(1,1)$*,* $(1,2)$*,* $(2,3)$*,* $(3,5)$*;*

(iii) for every $m \in \mathbb{N}$*,* $\operatorname{conv}(S_A)$ *is not unimodularly equivalent to* $[0,m]^2$*.*

In particular the $d = 2$ *instance of Conjecture 3 of [3] — hence the conjecture, which asserts all dimensions — is false.*

Proof. (i) The six pairwise differences are $(1,0)$, $(0,1)$, $(3,5)$, $(-1,1)$, $(2,5)$, $(3,4)$, each with coprime coordinates; apply Lemma 2.2.

(ii) Let $Q = \{(x,y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0,\ 5x - 2y \leq 5,\ 3y - 4x \leq 3\}$. Each of the four halfplanes is convex and contains the four points of S_A (sixteen evaluations; each bounding line passes through exactly two points of S_A, namely the edge pairs $\{(0,1),(0,0)\}$, $\{(0,0),(1,0)\}$, $\{(1,0),(3,5)\}$, $\{(0,1),(3,5)\}$), so $\operatorname{conv}(S_A) \subseteq Q$. For integer points of Q: the third and fourth inequalities give $15x - 15 \leq 6y \leq 8x + 6$, so $x \leq 3$, and then y is pinned down case by case — $x = 0$ forces $y \in \{0,1\}$; $x = 1$ forces $y \in \{0,1,2\}$; $x = 2$ forces $y = 3$; $x = 3$ forces $y = 5$. These are the seven listed points. Conversely all seven lie in $\operatorname{conv}(S_A)$: four are points of S_A, and

$$(1,1) = \tfrac{2}{5}(0,0) + \tfrac{2}{5}(1,0) + \tfrac{1}{5}(3,5), \qquad (1,2) = \tfrac{1}{3}\big[(0,0) + (0,1) + (3,5)\big],$$
$$(2,3) = \tfrac{1}{5}(0,0) + \tfrac{1}{5}(1,0) + \tfrac{3}{5}(3,5).$$

(iii) If $\Phi(x) = Ax + t$ with $A \in \mathrm{GL}(2,\mathbb{Z})$, $t \in \mathbb{Z}^2$, then Φ and Φ^{-1} map $\mathbb{Z}^2$ into $\mathbb{Z}^2$, so Φ restricts to a bijection $P \cap \mathbb{Z}^2 \to \Phi(P) \cap \mathbb{Z}^2$ for every $P \subseteq \mathbb{R}^2$: the lattice-point count is invariant under unimodular equivalence. But $\big|[0,m]^2 \cap \mathbb{Z}^2\big| = (m+1)^2$ is a perfect square, and $2^2 = 4 < 7 < 9 = 3^2$, so no m gives 7. With (ii), $\operatorname{conv}(S_A)$ is equivalent to no $[0,m]^2$.

Finally, (i) and (iii) contradict the "only if" direction of the quoted conjecture for $S = S_A$ — under either reading of "for any m", since (iii) denies the equivalence for every m individually. □

Remark 2.4. S_A is not the lattice-point set of its hull: it omits $(1,1)$, $(1,2)$, $(2,3)$. Conjecture 3 of [3] is stated for arbitrary bounded $S \subset \mathbb{Z}^d$, so this is no defect of the counterexample — but it does suggest the repair of restricting the conjecture to sets with $S = \operatorname{conv}(S) \cap \mathbb{Z}^d$. Section 3 shows that this repair also fails.

3 The full-lattice-set repair also fails

The results of this section are verified in exact integer and rational arithmetic (Section 5) but are *not* part of the Lean development.

Theorem 3.1 (witness T, $d = 2$)**.** *Let* $T = \{(0,1), (1,0), (1,1), (2,2)\} \subset \mathbb{Z}^2$*. Then* $T = \operatorname{conv}(T) \cap \mathbb{Z}^2$*,* $\beta_{\mathbb{Z}}(T) = 4 = 2^2$*, and* $\operatorname{conv}(T)$ *— the triangle with vertices* $(0,1)$*,* $(1,0)$*,* $(2,2)$*, of area* $3/2$ *— is unimodularly equivalent to no square* $[0,m]^2$*.*

Proof. The six pairwise differences $(1,-1)$, $(1,0)$, $(2,1)$, $(0,1)$, $(1,2)$, $(1,1)$ are primitive, so $\beta_{\mathbb{Z}}(T) = 4$ by Lemma 2.2. Since $(1,1) = \frac{1}{3}\big[(0,1) + (1,0) + (2,2)\big]$ is the centroid of the other three points, $\operatorname{conv}(T)$ is the triangle with the stated vertices; its area is $3/2$ by the shoelace formula. The triangle is cut out by $x + y \geq 1$, $2x - y \leq 2$, $2y - x \leq 2$ (each line through two vertices, each inequality satisfied by all of T and strictly by the centroid $(1,1)$), and a four-line case check gives the integer solutions: the second and third inequalities force $4x - 4 \leq 2 + x$, i.e. $x \le 2$, and $x \ge 0$ (for $x \le -1$ they give $2y \le 1$ and $y \ge 1 - x \ge 2$, impossible); then $x = 0 \Rightarrow y = 1$, $x = 1 \Rightarrow y \in \{0, 1\}$, $x = 2 \Rightarrow y = 2$. So $\mathrm{conv}(T) \cap \mathbb{Z}^2 = T$, with 4 points. If $\mathrm{conv}(T) = A\,[0, m]^2 + t$ were a unimodular image of a square, the count invariant of Theorem 2.3(iii) forces $(m + 1)^2 = 4$, i.e. $m = 1$; but $|\det A| = 1$, so unimodular maps preserve area, and $\mathrm{area}(\mathrm{conv}(T)) = 3/2 \ne 1 = \mathrm{area}([0, 1]^2)$. (Alternatively: an affine bijection preserves the number of vertices, and a triangle has $3 \ne 4$.) □

Theorem 3.2 (witness C, $d = 3$)**.** *Let*

$$C = \{(0, 2, 2),\ (1, 0, 1),\ (1, 1, 2),\ (1, 2, 2),\ (2, 1, 1),\ (2, 1, 2),\ (2, 2, 1),\ (3, 1, 3)\} \subset \mathbb{Z}^3.$$

Then $C = \mathrm{conv}(C) \cap \mathbb{Z}^3$*,* $\beta_{\mathbb{Z}}(C) = 8 = 2^3$*, and* $\mathrm{conv}(C)$*, of volume* $5/2$*, is unimodularly equivalent to no cube* $[0, m]^3$*.*

Proof. All 28 pairwise differences are primitive (28 exact gcd computations), so $\beta_{\mathbb{Z}}(C) = 8$ by Lemma 2.2. For $C = \mathrm{conv}(C) \cap \mathbb{Z}^3$ we use an exact supporting-halfspace certificate: collect every plane spanned by an affinely independent triple of points of C having all eight points on one (closed) side. Since $\mathrm{conv}(C)$ is full-dimensional (verified: four of the points are affinely independent), every facet plane of $\mathrm{conv}(C)$ is spanned by points of C and hence occurs in this collection, so the intersection of the collected halfspaces is exactly $\mathrm{conv}(C)$; enumerating the integer points of the coordinate bounding box $[0, 3] \times [0, 2] \times [1, 3]$ (which contains $\mathrm{conv}(C)$) against these inequalities returns precisely the eight points of C. The volume, $5/2$, is computed exactly from the same facet description by fan triangulation.

Now suppose $\mathrm{conv}(C) = A\,[0, m]^3 + t$ with $A \in \mathrm{GL}(3, \mathbb{Z})$, $t \in \mathbb{Z}^3$. The count invariant (as in Theorem 2.3(iii), verbatim in $d = 3$) gives $(m + 1)^3 = |C| = 8$, so $m = 1$: indeed $m = 0$ gives 1 point and $m \ge 2$ gives at least $27 > 8$. For $m = 1$ the map $\Phi(x) = Ax + t$ restricts to a bijection from $[0, 1]^3 \cap \mathbb{Z}^3 = \{0, 1\}^3$ onto $\mathrm{conv}(C) \cap \mathbb{Z}^3 = C$. For a finite $X \subset \mathbb{Z}^3$ let

$$N(X) \ = \ \#\{\ \{\{p, q\}, \{r, s\}\} : \{p, q\} \ne \{r, s\} \text{ pairs of distinct points of } X,\ p + q = r + s\ \},$$

the number of unordered coincidences among the pairwise sums. Since A is invertible, $p+q = r+s$ iff $\Phi(p) + \Phi(q) = \Phi(r) + \Phi(s)$, so N is preserved by Φ. But $N(\{0, 1\}^3) = 12$ — the $\binom{4}{2} = 6$ coincidences among the four antipodal vertex pairs, which all sum to $(1, 1, 1)$, plus one coincidence of the two diagonals in each of the 6 facets — while $N(C) = 2$: the only coinciding pair-sums in C are

$$\begin{aligned} (0, 2, 2) + (2, 1, 2) \ &= \ (2, 3, 4) \ = \ (1, 1, 2) + (1, 2, 2), \\ (1, 1, 2) + (2, 2, 1) \ &= \ (3, 3, 3) \ = \ (1, 2, 2) + (2, 1, 1). \end{aligned}$$

This contradiction rules out $m = 1$. (Cross-check, independent of N: unimodular maps preserve volume, and $\mathrm{vol}(\mathrm{conv}(C)) = 5/2 \ne 1 = \mathrm{vol}([0, 1]^3)$.) □

Remark 3.3 (what survives). All three witnesses exploit the regime $\mathrm{diam}_{\mathbb{Z}}(S) = 1$, where Lemma 2.2 collapses $\beta_{\mathbb{Z}}$ to the cardinality. We do not determine the status of the conjecture under the combined repair "$S = \mathrm{conv}(S) \cap \mathbb{Z}^d$ *and* $\mathrm{diam}_{\mathbb{Z}}(S) \ge 2$", which our witnesses do not reach; for $d = 2$ the Bárány–Füredi characterization [1, Theorem 2(iii)] cited by the authors is a natural route to a positive result there. We note only that the needed diameter restriction is somewhat awkward for the conjecture's motivation, since the sharpness example $\{0, 1\}^d$ of [3, Theorem 4.8] itself has lattice diameter 1. The "if" direction of the conjecture is not addressed by this note.

4 The Lean 4 artifact

Theorem 2.3 is formalized in Lean 4 [6] (toolchain `leanprover/lean4:v4.30.0`) over mathlib [5] (release tag `v4.30.0`), in a development of about 1100 lines. The library consists of seven files:

File	Content	Lines
`Borsuk/Defs.lean`	frozen definitions + faithfulness docstrings	282
`Borsuk/SegmentGcd.lean`	Lemma 2.1: segment count = gcd +1	252
`Borsuk/WitnessA.lean`	Lemma 2.2; $\mathrm{diam}_{\mathbb{Z}}(S_A) = 1$; $\beta_{\mathbb{Z}}(S_A) = 4$	148
`Borsuk/HullSeven.lean`	H-representation; $\lvert\operatorname{conv}(S_A) \cap \mathbb{Z}^2\rvert = 7$	176
`Borsuk/Unimodular.lean`	count invariance; $(m + 1)^2$ cube count; $(m + 1)^2 \neq 7$	167
`Borsuk/Main.lean`	assembly of the three main theorems	59
`Borsuk/Smoke.lean`	toolchain smoke tests	44

The three top-level results, verbatim from `Borsuk/Main.lean`:

```
theorem witnessA_kills_conjecture3 :
    betaZ SA = 4 ∧
      ∀ m : ℕ, ¬ UnimodEquiv (convexHull ℝ (toR '' (SA : Set Z2))) (cube m)

theorem conjecture3_counterexample :
    ∃ S : Finset Z2, betaZ S = 2 ^ 2 ∧
      ∀ m : ℕ, ¬ UnimodEquiv (convexHull ℝ (toR '' (S : Set Z2))) (cube m)

theorem conjecture3_false : ¬ Conjecture3For2D
```

where the formal statement of (the $d = 2$ instance of) the conjecture, from `Borsuk/Defs.lean`, is

```
def Conjecture3For2D : Prop :=
  ∀ S : Finset Z2,
    (betaZ S = 2 ^ 2 ↔
      ∃ m : ℕ, UnimodEquiv (convexHull ℝ (toR '' (S : Set Z2))) (cube m))
```

Encoding choices (faithfulness). The Lean kernel verifies the proofs; whether the formal *definitions* say what the paper says is the one link a proof checker cannot certify, so `Defs.lean` documents every definition against the verbatim LaTeX source of [3], with the following choices. $\mathbb{Z}^2$ is `Z2 = ℤ × ℤ`, embedded in $\mathbb{R}^2$ by the injection `toR`. A "bounded set $S \subset \mathbb{Z}^d$" is a `Finset Z2` (bounded subsets of $\mathbb{Z}^2$ are exactly the finite ones; the witness, a 4-element set, is bounded under any reading). f is `latticeLength`, the cardinality of $\{p \in \mathbb{Z}^2 : p \in [x, y]\}$ minus one, with mathlib's `segment` proved equal to the paper's $\operatorname{conv}(\{x, y\})$; $\mathrm{diam}_{\mathbb{Z}}$ is a double `Finset.sup`. Partitions are mathlib's `Finpartition` (pairwise-disjoint nonempty parts with union S), and `betaZ` is the infimum (`sInf`) of the achievable part counts of partitions all of whose parts have strictly smaller `latticeDiam` — the paper's "minimal size of a partition ... [with] strictly smaller lattice diameter". Unimodular equivalence is an explicit structure `UnimodMap`: four integer matrix entries with $ad - bc = \pm 1$ (the units of $\mathbb{Z}$) plus an integer translation, acting on $\mathbb{R}^2$, with $P = \varphi(Q)$ as the paper's $P = AQ + t$; the formal development proves that such maps biject $\mathbb{Z}^2$ (the paper's "lattice preserving maps"). `cube m` is $[0, m]^2 \subseteq \mathbb{R}^2$, and `latticeCount` is the number of lattice points. Two deliberate robustness margins: "for any $m \in \mathbb{N}$" is rendered as $\exists\, m$, but the proof establishes $\forall m,\ \neg\,$equiv, which negates the right-hand side under both the existential and the literal universal reading, for $\mathbb{N}$ with or without 0; and `UnimodEquiv` is stated for arbitrary subsets of $\mathbb{R}^2$ rather than lattice polytopes only — on the objects in the conjecture ($\mathrm{conv}(S_A)$ and $[0,m]^2$, both lattice polytopes) the readings coincide. One degenerate corner is flagged rather than hidden: for the empty set the empty partition is admissible and gives $\beta_{\mathbb{Z}}(\emptyset) = 0$, and for singleton S no admissible partition exists, so the achievable-count set is empty and the convention `sInf` $\emptyset = 0$ gives $\beta_{\mathbb{Z}}(S) = 0$ where the paper's minimum is simply undefined; if moreover $0 \in \mathbb{N}$, a one-point hull *is* unimodularly equivalent to the degenerate cube $[0,0]^2$, so a singleton by itself already falsifies the formal biconditional. The development makes no use of this junk route: the proof of `conjecture3_false` instantiates the conjecture only at $S = S_A$, with honest lattice diameter 1 and an honestly attained minimum $\beta_{\mathbb{Z}}(S_A) = 4$, and Theorem 2.3 refutes the "only if" direction in a form untouched by any convention for the degenerate cases.

Verification status. On 2026-06-11 and again on 2026-06-12 the project built cleanly: `lake build` reports `Build completed successfully (8483 jobs)`, exit code 0, no warnings. There are no `sorry` or `admit` terms, no `axiom` declarations, no `native_decide` (all decidability checks are kernel-checked `decide`), no `unsafe` or `partial` definitions, and no custom elaboration. The axiom audit (`lake env lean scripts/CheckAxioms.lean`) prints, for all sixteen audited theorems — including the three above and the supporting lemmas `ncard_segLatticePts`, `latticeDiam_SA`, `betaZ_eq_card_of_latticeDiam_eq_one`, `betaZ_SA`, `mem_hullSA_iff`, `latticeCount_hullSA`, `UnimodEquiv.latticeCount_eq`, `latticeCount_cube`, `succ_sq_ne_seven` — an axiom set contained in

[`propext`, `Classical.choice`, `Quot.sound`],

the three standard mathlib axioms; in particular no `sorryAx`. For example (one line per theorem, quoted exactly):

```
'Borsuk.witnessA_kills_conjecture3' depends on axioms: [propext, Classical.choice, Quot.sound]
'Borsuk.conjecture3_false' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Building. With elan installed, from the `lean/` directory of the artifact:

```
elan toolchain install leanprover/lean4:v4.30.0
lake exe cache get      # fetch mathlib v4.30.0 build cache (~1.6 GB)
lake build              # expect: Build completed successfully, exit 0
lake env lean scripts/CheckAxioms.lean   # axiom audit, output as above
```

The toolchain and mathlib revisions are pinned by `lean-toolchain`, `lakefile.toml`, and `lake-manifest.json`, so the build is reproducible; without the cache, mathlib builds from source in a few hours.

5 Verification of the numbers in this note

Every quantity quoted above is mechanically checkable. For Theorem 2.3 the authority is the Lean development of Section 4. For Theorems 3.1 and 3.2, and as an independent cross-check of the witness-A data, the script `_recompute.py` distributed with this note (Python standard library only; integer and `fractions.Fraction` arithmetic on every accept path, no floating point anywhere) re-verifies by direct recomputation: the $6+6+28$ primitivity checks; the H-representation of $\mathrm{conv}(S_A)$ (halfspace containment and the edge–vertex incidences), the enumeration of its 7 lattice points, and exact convex-combination certificates for the three non-elements of S_A; the triangle $\mathrm{conv}(T)$, its area $3/2$, and $\mathrm{conv}(T) \cap \mathbb{Z}^2 = T$; the supporting-halfspace certificate for $\mathrm{conv}(C) \cap \mathbb{Z}^3 = C$ (these halfspaces the script does derive from scratch), the exact volume $5/2$, the pair-sum invariants $N(C) = 2$ and $N(\{0,1\}^3) = 12$, and the cube counts $(m+1)^3$. It was re-run on 2026-06-12 with verdict `ALL CHECKS PASS` (27 checks). A second script, `_referee_recheck.py`, written during the hostile-referee pass of 2026-06-12 deliberately by different routes — hull membership by exact Carathéodory solves over $\mathbb{Q}$ instead of H-representations, vertices by the "not in the hull of the others" criterion, the volume by a divergence-style signed surface sum, and a full census of coinciding pair-sums — reaches the same conclusions (37 checks, `ALL CHECKS PASS`). The discovery pipeline had already verified witnesses A–C in exact arithmetic (three independent recomputations, plus the second-model-family re-verification noted in Section 1); both scripts above were written separately from it.

Data availability. The Lean project (source, pinned manifests, axiom-audit script), the two exact-arithmetic verification scripts, the discovery-time verification records, and the frozen copy of the arXiv v1 source of [3] against which the definitions were checked will be made available at `https://github.com/demonstrandum-research/[REPO]`.

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